PROTOTYPE DESCRIPTION

Smart Safety Companion

Overview

The Smart Safety Companion is a mobile application prototype created using MIT App Inventor. It assists users during emergency situations by detecting intense phone movements and instantly generating alerts that include the user's location and directional information.

1. Purpose of the Prototype

The primary objective of this prototype is to enhance personal safety. The application performs the following functions:

- Continuously monitors device motion
- Detects sudden or forceful shaking
- Tracks the user's GPS location
- Determines the device's direction
- Generates emergency alert messages
- Stores emergency contact numbers

Overall, the app acts as an automatic safety assistant that responds quickly during potentially dangerous situations.

2. User Interface Description

Application Title

The title "Smart Safety Companion" appears at the top of the screen to clearly identify the application.

Emergency Contact Section

- Users can add emergency contact numbers
- Contacts are stored using Tiny DB
- Saved data remains even after the app is closed
- These numbers are used to send alerts during emergencies

Activate SOS

- Initiates the monitoring process
- Activates accelerometer, GPS, and orientation sensors
- Updates the status to **"**Activated**"**
- Begins tracking device movements

Deactivate SOS

- Stops all monitoring operations
- Deactivates all sensors
- Resets collected values
- Changes the status to **"**Deactivated**"**

3.Sensor and Status Display

The application displays real-time information on the screen, including:

- Status – Indicates whether the app is Stopped, Monitoring, or Emergency Detected
- Accelerometer Values (X, Y, Z) – Shows device motion data
- Shake Value – Calculated using |X| + |Y| + |Z|
- Location – Displays latitude and longitude
- Direction – Shows orientation (North, East, South, or West)
- Alert Message – Presents emergency details such as time, location, and direction

4.Background Design

A technology-themed background is incorporated to enhance the visual appeal and improve the user experience.

5.Block Programming Logic

The application is developed using block-based programming in MIT App Inventor.

Global Variables

- monitoring – Indicates whether monitoring is active
- cooldown – Prevents repeated alerts within a short time
- shake Threshold – Minimum shake value required to trigger an emergency
- direction Text – Stores directional data
- alert Message – Stores the generated alert message

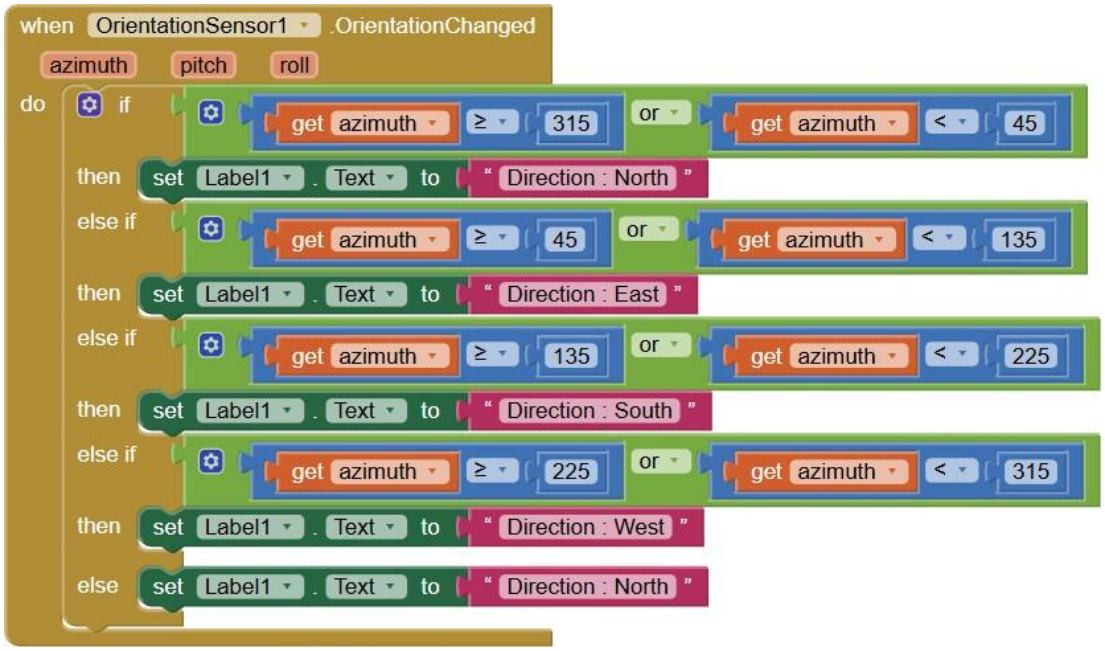
Working of Sensors

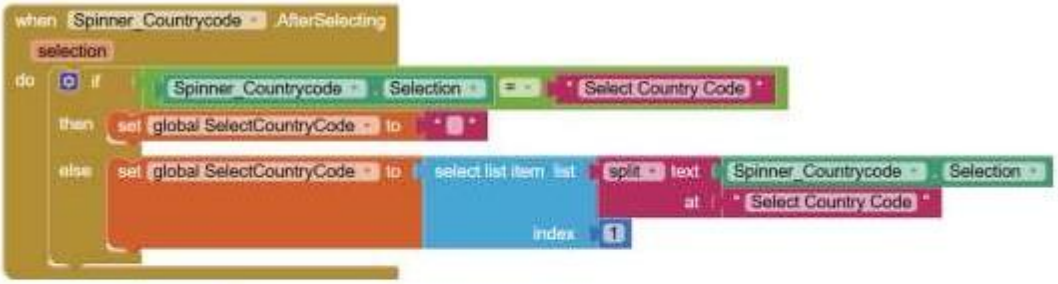
- The accelerometer tracks phone movement
- Shake value is calculated using X, Y, and Z coordinates
- When the shake value exceeds the defined threshold:
- An emergency is detected
- Location and direction data are captured
- An alert message is generated
- Cooldown is activated to avoid repeated notifications

Direction Detection

The orientation sensor determines the device's direction:

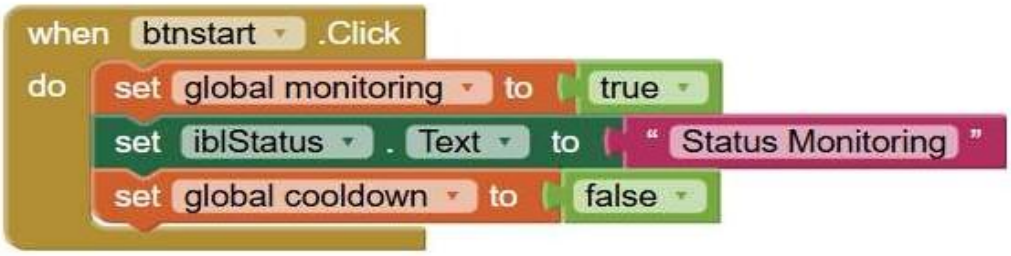
- 315° – 45° → North
- 45° – 135° → East
- 135° – 225° → South
- 225° – 315° → West

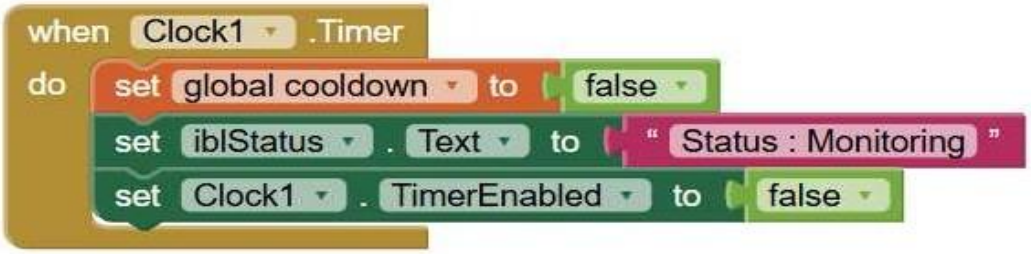


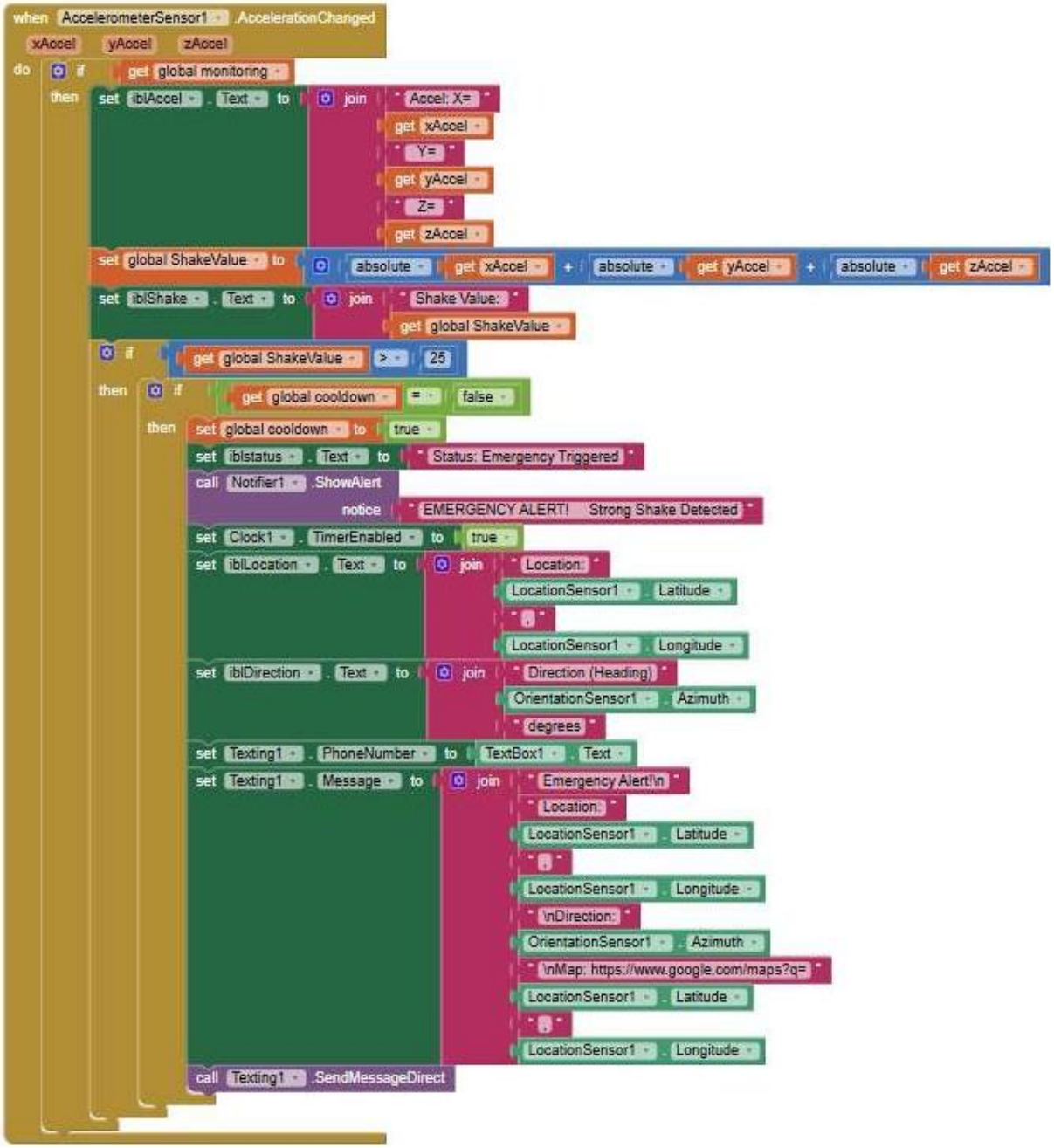

when Spinner_Countrycode .AfterSelecting
selection
do if Spinner_Countrycode . Selection = " Select Country Code "
then set global SelectCountryCode to " "
else set global SelectCountryCode to select list item list split text Spinner_Countrycode . Selection
at " Select Country Code "
index 1

when btnstop .Click
do set global monitoring to false
set lblStatus . Text to " Status Monitoring "
set global cooldown to false

initialize global cooldown to false
initialize global FullPhoneNumber to " "
initialize global ShakeValue to 0
initialize global SelectCountryCode to " "
initialize global monitoring to false
initialize global IsMonitoring to false


when btnstart .Click
do set global monitoring to true
set lblStatus . Text to " Status Monitoring "
set global cooldown to false


when Clock1 .Timer
do set global cooldown to false
set lblStatus . Text to " Status : Monitoring "
set Clock1 . TimerEnabled to false


when AccelerometerSensor1 .AccelerationChanged
xAccel yAccel zAccel
do if get global monitoring
then set lblAccel . Text to join "Accel: X=" get xAccel " Y=" get yAccel " Z=" get zAccel
set global ShakeValue to absolute get xAccel + absolute get yAccel + absolute get zAccel
set lblShake . Text to join "Shake Value:" get global ShakeValue
if get global ShakeValue > 25
then if get global cooldown = false
then set global cooldown to true
set lblstatus . Text to "Status: Emergency Triggered"
call Notifier1 .ShowAlert
notice "EMERGENCY ALERT! Strong Shake Detected"
set Clock1 . TimerEnabled to true
set lblLocation . Text to join "Location:" LocationSensor1 . Latitude "," LocationSensor1 . Longitude
set lblDirection . Text to join "Direction (Heading)" OrientationSensor1 . Azimuth "degrees"
set Texting1 . PhoneNumber to TextBox1 . Text
set Texting1 . Message to join "Emergency Alert!\n" "Location:" LocationSensor1 . Latitude "," LocationSensor1 . Longitude "\nDirection:" OrientationSensor1 . Azimuth "\nMap: https://www.google.com/maps?q=" LocationSensor1 . Latitude "," LocationSensor1 . Longitude
call Texting1 .SendMessageDirect

6.Overall Working

- The user enters emergency contacts
- Monitoring is started
- The app continuously observes device movement
- If strong shaking is detected:
- An emergency alert is displayed
- Location and direction are shown
- Monitoring can be stopped at any time

7.Conclusion

The Smart Safety Companion is a straightforward and effective safety application that leverages smartphone sensors to detect emergencies.

By integrating motion detection, GPS tracking, and directional information, it provides timely assistance during risky situations.

This prototype can be further expanded for applications such as:

- Women’s safety
- Accident detection
- Elderly care
- Emergency response systems